ME-214 Computer Programming & Applications

Complex Engineering Problem (CEP)

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Syeda Umayma Danish

ME 24008

SECTION A

Solar Radiation Analysis on a Flat Plate Collector using MATLAB

INTRODUCTION

A flat plate collector is a widely used solar energy device designed to capture and convert solar radiation into thermal energy. To maximize its efficiency, understanding the variation of solar radiation on the collector surface throughout the day is crucial. Factors such as the angle of incidence, solar altitude, air mass ratio, and atmospheric conditions significantly influence the total radiation received by the collector.

This project focuses on a MATLAB program designed to simulate the daily solar radiation received by a flat plate collector. To optimize collector performance, the program analyzes hourly variations influenced by factors like sun angles and atmospheric conditions. Users input the specific day of the year (n), using latitude and longitude coordinates obtained from Google Maps to ensure location-specific accuracy for any region.

The key solar parameters and radiative components considered in this analysis include:

- **Declination angle (δ):** Determines the tilt of the Earth's axis relative to the sun.
- **Hour angle (ω):** Measures the sun's position relative to solar noon.
- **Angle of incidence (θ):** Angle between the sun's rays and the collector surface.
- **Solar altitude (αs):** Vertical position of the sun in the sky.
- **Solar zenith (θ_x):** Angular position of the sun in the sky.
- **Air Mass Ratio (m):** Measure of how much atmosphere sunlight passes through relative to the zenith path.
- **Apparent Extraterrestrial Flux (A):** Intensity of solar radiation available just outside Earth's atmosphere (varies with Earth-Sun distance).
- **Optical Depth (k):** Parameter quantifying the atmosphere's blocking effect (scattering/absorption) on direct sunlight.
- **Direct Beam Radiation (IB):** Solar radiation arriving directly from the sun without being scattered by the atmosphere.
- **Diffuse Radiation (IDC):** Sunlight scattered by the atmosphere arriving indirectly from the entire sky dome onto the collector.
- **Reflected Radiation (IRC):** Sunlight reaching the collector after being reflected off the ground and surroundings.
- **Total Insolation on Collector (IC):** Sum of direct beam, diffuse sky, and ground-reflected radiation incident on the collector surface.

By performing hourly computations throughout the day, the MATLAB script generates a graph of total insolation on the collector. This output provides valuable insights into the optimal time frame for collector performance, based on assumed clear sky conditions and a ground reflectance (ρ) of 0.2.

ASSUMPTIONS

This simulation models the solar radiation incident on the flat plate collector based on several key assumptions:

- The calculations assume clear sky conditions, neglecting the effects of clouds and other atmospheric disturbances.
- A constant ground reflectance (ρ) of 0.2 is used to estimate the contribution of reflected radiation from the surroundings.
- The collector configuration is specifically defined as being tilted at an angle equal to the site's latitude (`slope = lat`) and oriented directly towards the equator (`azi = 0`), corresponding to a due South orientation in the Northern Hemisphere.

These assumptions define the specific scenario for which the hourly radiation values were calculated.

INPUTS

To ensure the simulation accurately reflects conditions for a specific location and time, the program requires key user inputs. These include:

- The precise geographical coordinates, latitude and longitude (obtained using Google Maps), which establish the site location.
- For the date input, the program prompts the user to select a month and enter the day. From this, it calculates the specific day of the year (represented as 'n') required for the solar radiation analysis.

This input methodology guarantees location-specific accuracy for the simulation across any region.

Calculations

The program performs calculations on an hourly basis (Local Time 0 to 23). Key parameters calculated include:

Solar Angles: Declination angle (δ), Hour angle (ω), Solar altitude (αs), Solar zenith (θz), Solar azimuth, and Angle of incidence (θ) on the tilted surface.
Time: Solar Time (ST) is calculated from Local Time (LT), adjusting for the Equation of Time (E) and longitude.
Atmospheric Parameters: Air Mass ratio (m), Apparent Extraterrestrial Flux (A), and Optical Depth (k) are calculated based on empirical clear-sky models.
Radiation Components: Using a clear-sky model and an assumed ground reflectance (albedo, p = 0.2), the following components incident on the collector are calculated:

- Direct Beam Radiation (IBC)
- Diffuse Sky Radiation (IDC)
- Ground-Reflected Radiation (IRC)

Total Insolation: The Total Incident Radiation on the Collector (IC) is computed as the sum: IC = IBC + IDC + IRC.

```
clc; clear;
format long

% Inputs
% (ϕ) latitude
lat = input('Enter Latitude (degrees): ');   % 24.947866475073255
while lat < -90 || lat > 90 || isnan(lat)
    error('Invalid latitude. Please enter a value between -90 and 90.');
end

% longitude
LL =  input('Enter Longitude (degrees): ');  % 67.1566328863201
while LL < -180 || LL > 180 || isnan(LL)
    error('Invalid longitude. Please enter a value between -180 and 180.');
end

month =
menu('MONTHS','January','February','March','April','May','June','July',...
    'August','September','October','November','December');
days_month = [31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31];

i = input('Day of Month: ');
while i < 1 || i > days_month(month) || isnan(i) || floor(i) ~= i
    error('Invalid day for selected month. Please enter a value between 1
and %d.\n', days_month(month));
end

n = sum(days_month(1:(month-1))) + i;
```

```
% Declination angle (δ)
dec_angle = 23.45 * sind(360 * ((284 + n) / 365));

% Time zone offset
GMT = ceil(LL / 15);
SL = GMT * 15;
B = 360 / 365 * (n - 1);
E = (229.2 / 60) * (0.000075 + 0.001868 * cosd(B) - 0.032077 * ...
    sind(B) - 0.014615 * cosd(2*B) - 0.04089 * sind(2*B));

slope = lat; % Tilt angle (B)
azi = 0; % Surface azimuth angle (γ)
```

```
% Pre-allocate arrays for hourly values
% Initialize with NaN for angles/ratios, zeros for radiation/time
```

```
hours_of_day = 0:23;
ST_values = NaN(1, 24);          % Solar Time
w_values = NaN(1, 24);           % Hour Angle (deg)
as_values = NaN(1, 24);          % Solar Altitude (deg)
sz_values = NaN(1, 24);          % Solar Zenith (deg)
s_az_values = NaN(1, 24);        % Solar Azimuth (deg)
ang_inc_values = NaN(1, 24);     % Angle of Incidence (deg)
m_values = NaN(1, 24);           % Air Mass ratio
IB_values = zeros(1, 24);        % Direct Beam Radiation (W/m^2) - DNI
IBC_values = zeros(1, 24);       % Beam Radiation on collector (W/m^2)
IDC_values = zeros(1, 24);       % Diffuse Radiation on collector (W/m^2)
IRC_values = zeros(1, 24);       % Reflected Radiation on collector (W/m^2)
IC_values = zeros(1, 24);        % Total Radiation on the collector (W/m^2)
```

```
for LT = 0:23 % Local Time
    idx = LT + 1; % Array index (1 to 24)

    ST = (E - (4 * (SL - LL)) / 60 ) + LT; % Solar time
    w = (ST - 12) * 15; % Hour angle (ω)

   % Store scalar values in arrays
    ST_values(idx) = ST;
    w_values(idx) = w;


    % solar altitude angle
    as = asind(cosd(lat) * cosd(dec_angle) * cosd(w) + sind(lat) *
sind(dec_angle));
   as_values(idx) = as;
    % solar zenith angle
    sz = acosd(cosd(lat) * cosd(dec_angle) * cosd(w) + sind(lat) *
sind(dec_angle));
    sz_values(idx) = sz;
    % angle of incidence (θ)
    ang_inc = acosd(sind(dec_angle) * sind(lat) * cosd(slope) -
sind(dec_angle) * cosd(lat) * sind(slope) * cosd(azi) + ...
        cosd(dec_angle) * cosd(lat) * cosd(slope) * cosd(w) +
cosd(dec_angle) * sind(lat) * sind(slope) * cosd(azi) * cosd(w) + ...
        cosd(dec_angle) * sind(slope) * sind(azi) * sind(w) );
    ang_inc_values(idx) = ang_inc;
    if as > 0
        s_az = sign(w) * abs(acosd((cosd(sz) * sind(lat) -
sind(dec_angle)) / (sind(sz) * cosd(lat))));
        s_az_values(idx) = s_az;
        % Air-Mass ratio (m)
        m = 1 / sind(as);
        m_values(idx) = m;
        % W/m2 (Apparent extraterrestrial flux)
```

```
        A = 1160 + 75 * sind((360 / 365 )*(n - 275));

        % Optical Depth
        k = 0.174 + 0.035 * sind((360 / 365) * (n - 100));

        % Direct Beam Solar Radiation
        IB = A * exp(-k * m);
        IB_values(idx) = IB;
        % Beam Radiation on collector
        IBC = IB * cosd(ang_inc);
        IBC_values(idx) = IBC;
        % Sky Diffuse Factor (C)
        C = 0.095 + 0.04 * sind((360 / 365) * (n - 100));

        % Diffuse radiation on collector
        IDC = C * IB * ((1 + cosd(slope)) / 2 );
        IDC_values(idx) = IDC;
        p = 0.2; % Ground reflectance

        % Reflected Radiation on the collector (IRC)
        IRC = p * IB * (sind(as) + C) * ((1 - cosd(slope)) / 2);
        IRC_values(idx) = IRC;
        % Total Radiation on the collector (IC)
        IC = IBC + IDC + IRC;
        IC_values(idx) = IC;
    else
        IC_values(idx) = 0 ;
    end
end
```

```
% Making sure all data are column vectors for the table

hours_col = hours_of_day(:); % Hour (0-23)
ST_col = ST_values(:);
w_col = w_values(:);
as_col = as_values(:);
sz_col = sz_values(:);
s_az_col = s_az_values(:);
ang_inc_col = ang_inc_values(:);
m_col = m_values(:);
IB_col = IB_values(:);
IBC_col = IBC_values(:);
IDC_col = IDC_values(:);
IRC_col = IRC_values(:);
IC_col = IC_values(:);
```

```
% Create the table
hourlyTable = table(hours_col, ST_col, w_col, as_col, sz_col, s_az_col,
ang_inc_col, m_col, IB_col, IBC_col, IDC_col, IRC_col, IC_col, ...
    'VariableNames', {'Hour', 'SolarTime', 'HourAngle', 'Altitude',
'Zenith', 'SolarAzim', 'IncidAngle', 'AirMass', 'IB', 'IBC', 'IDC', 'IRC',
'IC_Total'});
% Display Hourly Data Table
fprintf('\n\nHOURLY DATA:\n');
```

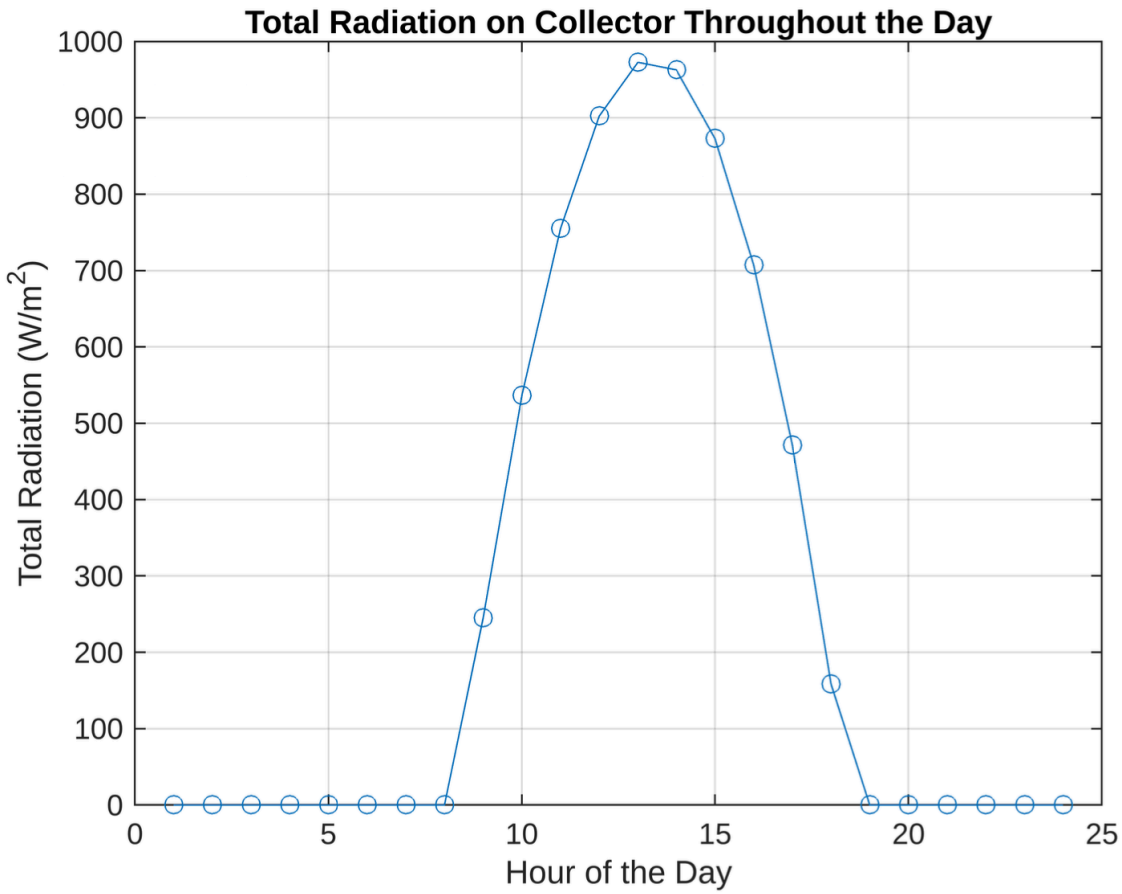
```
HOURLY DATA:
```

```
% Display the table
disp(hourlyTable);
```

Hour	SolarTime	HourAngle	Altitude	Zenith	SolarAzim	IncidAngle
0	-0.376176134832555	-185.642642022488	-84.3699664636537	174.369966463654	NaN	156.647042019817
1	0.623823865167445	-170.642642022488	-81.1515639031039	171.151563903104	NaN	155.54333199541
2	1.62382386516744	-155.642642022488	-67.6345745192453	157.634574519245	NaN	147.186101327834
3	2.62382386516744	-140.642642022488	-54.0493357435087	144.049335743509	NaN	135.505803405222
4	3.62382386516744	-125.642642022488	-40.5562305378248	130.556230537825	NaN	122.520139609571
5	4.62382386516745	-110.642642022488	-27.2354003118394	117.235400311839	NaN	108.97977289499
6	5.62382386516745	-95.6426420224883	-14.1821082269797	104.18210822698	NaN	95.2043658208947
7	6.62382386516745	-80.6426420224883	-1.53325111175061	91.5332511117506	NaN	81.3731289865286
8	7.62382386516745	-65.6426420224883	10.4992175382803	79.5007824617198	-58.7316680062665	67.6363117134137
9	8.62382386516745	-50.6426420224883	21.5733053539694	68.4266946460306	-50.0901864717131	54.1941629872409
10	9.62382386516745	-35.6426420224883	31.1319789071853	58.8680210928147	-38.9062837601461	41.4332899426773
11	10.6238238651674	-20.6426420224883	38.3165523753136	51.6834476246864	-24.4893386239552	30.3085240512455
12	11.6238238651674	-5.64264202248832	42.0405273143797	47.9594726856203	-7.01551435311875	23.3529579801827
13	12.6238238651674	9.35735797751168	41.4968352401173	48.5031647598827	11.5526194788306	24.4566680045896
14	13.6238238651674	24.3573579775117	36.8166166435191	53.1833833564809	28.3772032173637	32.8138986721657
15	14.6238238651674	39.3573579775117	28.9471591399267	61.0528408600733	41.9565391329467	44.4941965947779
16	15.6238238651674	54.3573579775117	18.9458245497332	71.0541754502669	52.4356146971091	57.4798603904293
17	16.6238238651674	69.3573579775117	7.59254090218344	82.4074590978166	60.5699671704471	71.02022710501
18	17.6238238651674	84.3573579775117	-4.61819014918136	94.6181901491813	NaN	84.7956341791054
19	18.6238238651674	99.3573579775117	-17.3833172312146	107.383317231215	NaN	98.6268710134714
20	19.6238238651674	114.357357977512	-30.5135500586766	120.513550058677	NaN	112.363688286586
21	20.6238238651674	129.357357977512	-43.8848550724909	133.884855072491	NaN	125.805837012759
22	21.6238238651674	144.357357977512	-57.4083795313682	147.408379531368	NaN	138.566710057323
23	22.6238238651674	159.357357977512	-71.0012614466919	161.001261446692	NaN	149.691475948754

AirMass	IB	IBC	IDC	IRC	IC_Total
NaN	0	0	0	0	0
NaN	0	0	0	0	0
NaN	0	0	0	0	0
NaN	0	0	0	0	0
NaN	0	0	0	0	0
NaN	0	0	0	0	0
NaN	0	0	0	0	0
NaN	0	0	0	0	0
5.48780863051838	554.680665696406	211.047318386864	32.5546631439645	1.26173818900941	244.863719719838
2.71967256954427	828.038627116566	484.43596611772	48.5982660710311	3.31653322925474	536.350765418006
1.93419421421071	927.740435239043	695.551782584941	54.4498469517102	5.00846458190835	755.01009411856
1.61288833863748	971.905561434988	839.065976867363	57.0419344264279	6.18091551592547	902.288826809716
1.4933038211052	988.874685885417	907.866457159159	58.0378662562099	6.7469558779353	972.651279293304
1.50925472368297	986.594230035626	898.071704910126	57.9040244322634	6.6662439722088	962.641973314598
1.66873646296405	964.080731567624	810.247355932429	56.5826887446425	5.94449380537411	872.774538482446
2.06610375495523	910.195179317457	649.261735594598	53.4201014934207	4.63341022415224	707.315247312171
3.08001272356598	785.957901580359	422.527847123639	46.1285137803778	2.83251500711206	471.488875911129
7.56845591177788	410.441847706459	133.48978268794	24.0891686309172	0.741786060945752	158.320737379803
NaN	0	0	0	0	0
NaN	0	0	0	0	0
NaN	0	0	0	0	0
NaN	0	0	0	0	0
NaN	0	0	0	0	0
NaN	0	0	0	0	0

```
% Plotting the Total Radiation
figure;
plot(1:24, IC_values, '-o'); % Plot IC_values against hours (1 to 24)
title('Total Radiation on Collector Throughout the Day');
xlabel('Hour of the Day');
ylabel('Total Radiation (W/m^2)');
grid on;
```



Results

The program outputs the results in two primary formats:

- Hourly Data Table
- Total Radiation Graph

The specific results presented in this report's hourly data table and total radiation graph correspond to the simulation being run for an input date of **December 6th**. This date determines the day of the year (n), which dictates key solar parameters for the calculation. The December 6th input yields the solar declination angle characteristic of early winter in the Northern Hemisphere (-22.5 degrees) and also sets the values for the Equation of Time and the seasonal atmospheric parameters (A, k, C) used in the clear-sky radiation model for that day.

Hourly Data Table:

A comprehensive table is generated, listing the calculated values for all key parameters over a 24-hour period (0-23 hours). This includes solar angles (Altitude, Zenith, Azimuth, Incidence Angle), time variables (Solar Time, Hour Angle), air mass ratio, and the various radiation components (IB, IBC, IDC, IRC, IC_Total).

The table clearly illustrates how things change through the day and night by showing periods with no solar radiation (night hours) and the variation during daylight hours, with radiation values peaking near solar noon.

- **During night hours (approx. 0-7 and 18-23),**

  Altitude is negative, and all radiation values (`IB` to `IC_Total`) are zero. Solar Azimuth and Air Mass are NaN as they aren't calculated when the sun is down.

- **During daylight hours (approx. 8-17),**

  The radiation values increase, peak around solar noon, when the sun is highest (maximum solar altitude) and hits the collector most directly (minimum angle of incidence), and then decrease towards sunset.

Graphical Representation:

A plot has been generated that displays the Total Radiation on the Collector (IC_Total in W/m^2) as it changes over the Hour of the Day. This graph visually represents how the Total Radiation on the Collector varies throughout the day, exhibiting a characteristic bell shape that peaks around midday, consistent with the data in the table.

> **X-axis:** Represents the end of each hour, from 1 (representing 00:00-01:00) to 24.
>
> **Y-axis:** Shows the 'Total Radiation (W/m^2)', which is the `IC_Total` value calculated for each hour.
>
> **The Curve:** The plotted line with circles shows how the total solar energy striking the collector surface changes hour by hour.

What the graph shows:

- The radiation starts low in the early morning hours.
- It rises steadily as the sun gets higher in the sky.
- The radiation reaches its maximum around midday (between hours 12 and 14, corresponding to roughly 11:00 AM to 1:00 PM local time),indicating the time when the collector receives the most solar energy under clear sky conditions. This peak matches the highest values in the `IC_Total` column of the table.
- In the afternoon, the radiation decreases as the sun gets lower.
- It returns to near zero in the evening and night hours

Conclusion

The developed MATLAB program successfully simulates the hourly solar radiation that would fall on a flat plate collector. This program requires the user to input a location and day, and it then simulates the radiation based on a collector tilted at the local latitude and oriented towards the equator. The simulation is based on standard solar geometry calculations and a specific clear-sky radiation model.

The program generates two key outputs: a detailed table showing radiation levels for each hour, and a summary graph illustrating the total solar energy received. These outputs offer both numerical and visual understanding of how the solar collector would perform throughout the day under ideal clear weather. This information can be helpful in determining the best times for energy harvesting with the given setup.

Rubrics for Complex Engineering Problem					
ME-214 Computer Programming and Applications					
Criterion	**Level of attainment**				
	Excellent (5)	**Good (4)**	**Fair (2)**	**Need Improvement (1)**	**Poor (0)**
Realization of Problems	Able to clearly comprehend the problem and develop a plan accordingly	Able to comprehend the problem and develop a plan with some deficiencies	Able to comprehend the problem and develop a plan with some deficiencies and inaccuracies	Able to comprehend the problem and develop a plan with major deficiencies and inaccuracies	Unable to comprehend the problem and develop a plan
Problem Solving Skills	Able to cover all aspects of the problem & properly carryout the calculations accurately	Able to cover many aspects of the problem & properly carryout the calculations accurately with minor deficiencies	Able to cover some aspects of the problem & properly carryout the calculations accurately with minor deficiencies and inaccuracies	Able to cover a few aspects of the problem & properly carryout the calculations accurately with major deficiencies and inaccuracies	Unable to cover any aspect of the problem
Usage of software	Able to use the software efficiently with effective programming techniques used throughout the program	Able to use the software sufficiently with effective programming techniques used in some parts of the program	Able to use the software sufficiently with effective programmin g techniques used in a few parts of the program	Able to use the software not sufficient enough to cater effective programming techniques	Unable to use software
Extraction of results and Presentation of work	Able to accomplish correct results the and present the results efficiently	Able to accomplish correct results and present the results sufficiently	Able to accomplish results with minor inaccuracies and present the results sufficiently with minor deficiencies	Able to accomplish results with major inaccuracies and present the results sufficiently with major deficiencies	Unable to accomplish the required results

Marks Obtained = _____ Teacher Signature: _____